1 MSDS: Materials Safety Data Sheets

1. standard documents that provide safety info on Materials
2. provided by suppliers
3. include info on:
   - composition
   - handling + PPE (Personal protective equipment)
   - hazards
   - shipping regulations

2 Legislation

- can support all 3 capitals
- includes state, national, and internation standards+agreements

3 EV case study

People:

- range anxiety
-

Planet:

- Concerns over the carbon emission of energy grid
- are carbon emissions better for EVs?
- legislation in EU requires battery recycling

Prosperity:

- Li only produced by 2 major countries
- current Li production does not meet anticipated demand
- Nd production too low
- 95% of Nd from one country(China)

4 Final presentation material index

Objective - Minimize deflection δ:

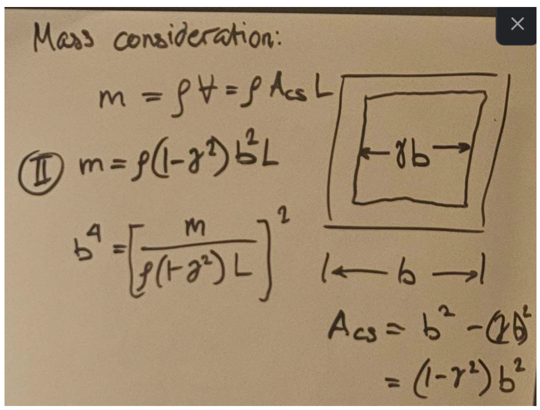
$$\delta = \frac{F_f L^3}{C_1 E I}$$

Constraints - Failure force F_f:

$$F_f = Z \frac{\sigma_y}{L}$$

Since F_f, L, and C_1 are constants, $\delta \propto EI$

Recover density term from Inertia:



From $m = \rho V$:

$$b^4 = \left[\frac{m}{\rho(1-\gamma^2)L}\right]^2$$

expanding Inertia term:

$$I = \frac{b^4}{12} - \frac{\gamma^4 b^4}{12}$$
$$I = \frac{(1-\gamma)^4}{12} b^4$$

combining inertia and mass equations:

$$I = \frac{(1-\gamma)^4}{12(1-\gamma^2)^2} \frac{m^2}{\rho^2 L^2}$$

Plug inertia term back into objective function:

$$\delta = \frac{F_f L^3}{C_1 E} \frac{12(1-\gamma^2)^2}{(1-\gamma)^4} \frac{\rho^2 L^2}{m^2} = \frac{F_f L^3}{C_1} \frac{12(1-\gamma^2)^2}{(1-\gamma)^4} \frac{L^2}{m^2} \left(\frac{\rho^2}{E}\right)$$

material index M:

$$M = \frac{\rho^2}{E}$$
$$\log M = 2\log\rho - \log E$$
$$\log E = 2\log\rho - \log M$$